Hybrid BLE and GPS Location Tagging System

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Introduction

Challenge: Combining two complimentary wireless technologies to achieve reliable, seamless location tagging

BLE: Fast, low-power proximity detection via RSSI, but limited range (5-30m) and vulnerable to signal interference.

GPS: Accurate global positioning (2.5-5m) with wide coverage, but slow update rates (~1Hz) and poor indoor performance.

Solution: A hybrid handshake system that uses BLE as the primary trigger and GPS as the validation layer.

Methodology

Approach: Iterative agile prototyping with a hybrid sensing model

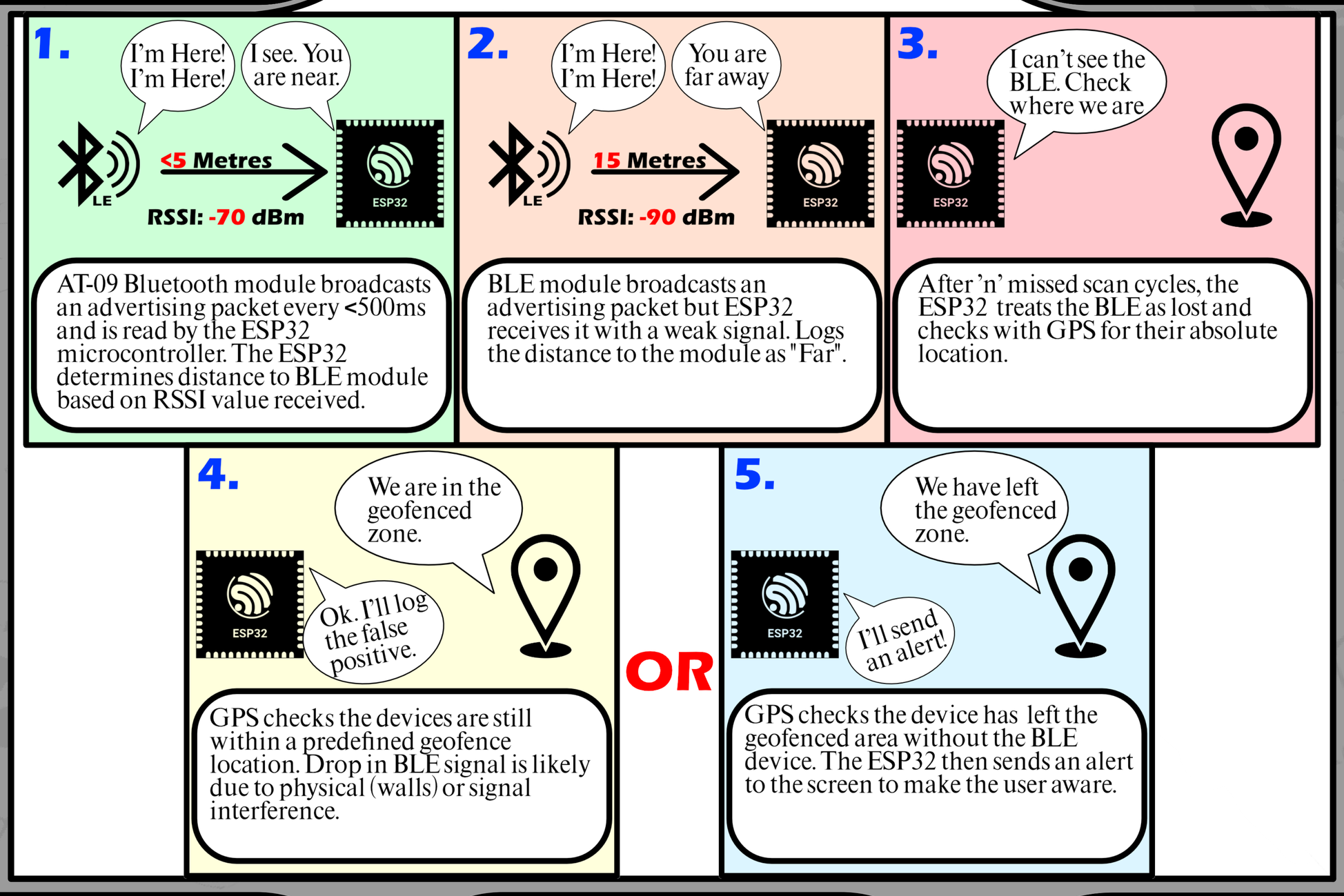
- BLE continuously monitors proximity using signal strength (RSSI)
- Loss of BLE signal triggers a GPS verification step
- GPS data is used to confirm true movement vs signal interference
- System classifies events (departure, false loss) based on both inputs
- Loss is either logged or acted upon accordingly

Aims & Objectives

Aim: Design and evaluate a hybrid BLE/GPS system for context aware location tagging.

Objectives:

- Implement BLE Proximity detection using RSSI values
- Integrate GPS for real world coordinate acquisition
- Develop a deterministic BLE-to-GPS handshake mechanism
- Classify events based on RSSI value (near, far, lost)
- Log system data for performance analysis



Technical Challenges

BLE Reliability:

- Signal interference / unstable RSSI values
- Inconsistent device identification (MAC)

GPS Integration:

- GPS module transmitted data but output was in an unknown format
- Limited ability to validate BLE events without reliable coordinates

System Impact:

- Reduced consistency of proximity tracking
- Incomplete automated handshake validation

Testing & Evaluation

BLE testing demonstrated successful proximity detection using **RSSI**, with fast response times achieved through scan intervals of **<500 ms**, and the detection logic was validated under controlled conditions.

In contrast, the GPS module consistently transmitted data but failed to produce readable output; however, when hard-coded coordinate values were used, the system was able to simulate GPS behaviour and validate the handshake logic.

Conclusion

Outcome: A hybrid BLE/GPS approach provides a strong foundation for seamless location tagging

- BLE enables fast, real-time proximity detection
- GPS provides positional context for validation
- The handshake model improves accuracy and reduces false detections

Validation:

- System logic and behaviour successfully demonstrated
- Hybrid sensing approach proven effective at a conceptual level

Future Work:

- Integrate fully functional GPS data for real-time validation
- Replace BLE hardware with a stable, identifiable device
- Refine thresholds and test in real-world environments
- Extend system for broader IoT and context-aware applications

References:

Chai, Song, Renbo An, and Zhengzhong Du. 'An Indoor Positioning Algorithm Using Bluetooth Low Energy RSSI'. April 2016, 274–76. https://doi.org/10.2991/amsee-16.2016.72

'Accuracy and Precision of GNSS in the Field'. In GPS and GNSS Technology in Geosciences. Elsevier, 2021. https://doi.org/10.1016/B978-0-12-818617-6.00002-0

Wolf, Jean. 'Defining GPS and Its Capabilities'. In Handbook of Transport Geography and Spatial Systems, vol. 5, edited by David A. Hensher, Kenneth J. Button, Kingsley E. Haynes, and Peter R. Stopher. Emerald Group Publishing Limited, 2004. https://doi.org/10.1108/9781615832538-023

Lee, Yi-Shiun, Yong-Yi Fanjiang, Chi-Huang Hung, and Yung-Shiang Huang. BLE Signal Processing and Machine Learning for Indoor Behavior Classification. 25, no. 14 (2025): 4496. https://doi.org/10.3390/s25144496

Bluetooth SIG Proprietary. 'Specification of the Bluetooth System'. Version BLUETOOTH SPECIFICATION Version 4.2. 2 December 2014. https://www.bluetooth.com/specifications/specs/core-specification-4-2/

Bisdikian, C. 'An Overview of the Bluetooth Wireless Technology'. IEEE Communications Magazine 39, no. 12 (2001): 86–94. https://doi.org/10.1109/35.968817

U-Blox. 'NEO-M8 Series'. 25 June 2015. https://www.u-blox.com/en/product/neo-m8-series

Mackensen, Elke, Matthias Lai, and Thomas M. Wendt. 'Bluetooth Low Energy (BLE) Based Wireless Sensors'. 2012 IEEE SENSORS, October 2012, 1–4. https://doi.org/10.1109/ICSENS.2012.6411303

Mackensen, Elke, Matthias Lai, and Thomas M. Wendt. 'Bluetooth Low Energy (BLE) Based Wireless Sensors'. 2012 IEEE SENSORS, October 2012, 1–4. https://doi.org/10.1109/ICSENS.2012.6411303Mackensen, Elke, Matthias Lai, and Thomas M. Wendt. 'Bluetooth Low Energy (BLE) Based Wireless Sensors'. *2012 IEEE SENSORS*, October 2012, 1–4. https://doi.org/10.1109/ICSENS.2012.6411303